

Q.1a)

Consider the above keypads and answer the questions below:

- 1) Why does an ATM keypad look different from a calculator keypad?

An ATM keypad may look different from a calculator keypad because it is designed for **different user tasks and contexts**.

HCI considers **user familiarity, mental models, consistency, and error prevention** when choosing the keypad layout.

- 2) Should the ATM use the telephone layout or calculator layout?

The ATM should use the **layout that is most familiar to its target users** and supports accurate number entry.

In HCI, **consistency with users' existing mental models** helps reduce confusion and input errors.

- 3) What problems may occur if we change the layout?

Changing a familiar layout can violate users' **mental models and expectations**.

This may cause **confusion, increased errors, and slower task completion**.

Q.1b)

A team designing a new e-learning platform skips user research, assuming that since the designers are young and tech-savvy, the target students (also young) will have the same expectations. After launch, many students struggle to navigate the platform. Using "user focus" concepts, explain

1. what went wrong and

The team assumed users had the same needs and expectations, ignoring the **User Focus** principle and actual user research.

2. what the team should have done differently

They should have studied the students' needs, goals, experience, and expectations before designing.

3. List one technique that could have identified the problem early on despite the mistake in part 1.

Usability testing

Q.2a) Read the scenario below. Identify every interaction paradigm

[08 points]

Ali's alarm goes off. He says, "Hey Google, stop," and it turns off. He picks up his phone and scrolls Instagram, pinching a photo to zoom in on it. He opens a design app to touch up the photo by dragging a filter slider, and the picture updates its brightness on screen exactly as it will look when exported.

While reading a news article, he taps a highlighted phrase, which jumps him to a linked video; from the video he taps through to the comments section, then to the creator's profile. On his laptop, he drags a file into the Trash icon to delete it, then resizes a window and switches between two open apps using the taskbar. He tells his AI assistant, "Book me a table for two at 8 pm at my usual place," and walks away while the assistant searches, finds availability, and confirms the booking on its own. As he leaves for work, the hallway lights turn on automatically as he walks through, and the thermostat has already adjusted the house temperature because it sensed he's about to leave; he never touched a switch or gave a command for either.

Note: No marks for listing paradigm names without tying each one to a specific moment and its defining feature; e.g., "WIMP" alone is not enough; you must say which WIMP element appears and where.

- **Voice Interaction:** Ali says “**Hey Google, stop**” to turn off the alarm. The system responds to a **spoken command**.
- **Direct Manipulation:** Ali **pinches the photo to zoom and drags the filter slider**. He directly manipulates on-screen objects and gets **immediate feedback**.
- **WYSIWYG:** The photo's brightness **updates on screen exactly as it will look when exported**. What Ali sees is what he will get.
- **Hypertext/Navigation:** Ali **taps a highlighted phrase → video → comments → creator profile**. He navigates through **linked information**.
- **WIMP:** Ali **drags a file to the Trash icon, resizes a window**, and uses the **taskbar** to switch apps. These demonstrate **windows, icons, menus, and pointer-based interaction**.
- **Agent/Intelligent Assistant:** Ali tells the AI “**Book me a table...**”, and it independently searches, finds availability, and confirms the booking. The system **acts autonomously on the user's behalf**.
- **Ubiquitous/Pervasive Computing:** The **hallway lights turn on automatically** and the **thermostat adjusts the temperature** based on sensing Ali's presence/activity. Computing is embedded in the environment and requires **no explicit interaction**.

Q2b) "The interface is not just about functions; it's about translation between the user and the system." [10 points]

Discuss this statement in the context of Human-Computer Interaction. In your answer, explain how an interface acts as a bridge between the user and the system by addressing the following:

1. Norman's Seven Stages (demonstrate understanding of the execution-evaluation loop)

Norman's Seven Stages describe how a user **forms a goal, plans and executes an action, then perceives and evaluates the system's response.**

The **execution phase** involves deciding and performing an action, while the **evaluation phase** involves perceiving, interpreting, and evaluating the system's feedback against the original goal.

2. Gulfs of Execution and Evaluation (explain where they occur in the interaction process, and discuss why minimizing these gulfs is important for usability.)

The Gulf of Execution is the gap between the user's goal and the actions available, while the Gulf of Evaluation is the gap between the system's response and the user's understanding of it.

3. At least **THREE** interaction styles (explain how each one translates between user and system differently)

Command-line translates user intentions through typed commands, **menu** selection through visible options, and **direct manipulation** through actions on visible objects.

4. Ergonomics (explain how physical design affects the translation process)

Ergonomics ensures that physical interface elements such as buttons, keyboards, and screens match users' physical capabilities and limitations.

5. Context (explain how social/organizational context influences interaction)

Social and organizational context affects interaction by influencing users' roles, tasks, access rights, and the way they use the system.